Explaining the better move

Nine rounds of hill-climbing a game review's lessons, graded on the board and by a blind judge

Chess Coach research lab

Abstract

At each key moment of a game review, the coaching app explains what the move played allowed, what the engine's better move does instead, and the difference between them. We measured those explanations on the 329 key moments of 60 of the player's games, with three grades checked against the board, and changed the code that writes them one idea per round, keeping a round only when no grade fell on either half of the games. Where a change wrote something the grades cannot check, a judge model compared the old and new texts blind, twice, with the order reversed. Over nine rounds (seven kept), the share of held-out moments whose texts make no false checkable claim rose from 90.1% to 100.0%, and the share that names a concrete consequence of the better move from 75.0% to 82.9%. The judge preferred every kept round's texts in nearly every comparison it decided. The kept code is the app's. The lab recomputes every published number from the records, and the app rerun on fresh engine evidence scores 100.0% and 82.9% on the same two measures.

1 What the player reads

When a game review stops at a key moment (a mistake, a blunder, a missed win), the app shows a card with a headline and a verdict, a lesson panel (what the move played allowed, what the better move does instead, the difference between the two lines, a takeaway), a Why sheet with the evidence behind the verdict, and a retry: the player may try the position again, with a hint and then the answer. Every one of those texts is written by heuristics in the app's review and knowledge services from the engine's evidence. No language model writes them, so every claim they make can be traced to a template, and most can be checked on the board.

The player is an adult club player, rated 948–1167 on chess.com across the 60 games studied here. Two things matter: that the texts are true, and that they say *why* the better move is better, in terms the player can use at the board, not only by how much.

2 Data and measures

Moments. The 329 key moments the review stops at in 60 of the player's games, from the app's phone-profile engine evidence for each game. The games are split 30 for train and 30 for test, the same split as the accuracy project, so no game is train in one and test in the other: 177 train moments and 152 test moments. A round was designed on train moments only.

Grades. Each moment's texts get three grades, all checked against the board with chess.js and against the stored engine lines (`grade.mjs`):

- **faithful**: 1 unless a checkable claim in the visible text is false: an illegal move, a line that does not replay, a piece said to attack what it does not, material said to be won that the line does not win, a mate that is not there, or an "if they take it with" answer that is not the engine's line or whose claim does not hold at its end.
- **concrete**: 1 when the better move's explanation names a consequence (material, a mate, a named threat, a prevented reply) and the text says what the move played allowed or missed.
- **named_deep**: 1 when every move the text names as best rests on a search at the app's best budget. It is a property of the evidence, not of the words, and is held, not targeted.

A fourth grade, **faithfulAll**, is faithful over the lesson the Why sheet also shows on a moment whose card shows only a comment. A clause no template recognises is reported as unparsed, never graded true. The grader was checked by hand on 19 moments before any round; the hand verdicts and the baseline's grades agree on all three grades for 19 of them.

Every rate below is a share of key moments, with a 95% interval from a bootstrap over games (a game's moments share its evidence, so the game is the unit). A variant's difference from the baseline is paired on the same moments.

The judge. The grades cannot see whether a new clause *helps*. For a round that writes something new, every moment whose texts change is shown to a judge model as two versions (X and Y, by a seeded coin), each laid out as the app lays it out, with the position, the move, the verdict, the engine's evaluation and best line. The judge answers which version would help the player more (understand why the better move is better, and know what to look for next time), or that they are the same, with one sentence of reason. It answers twice, with X and Y swapped the second time. The rule, fixed before any run: keep the round if, pooling both runs, the new texts are preferred in more than half of the decided comparisons on train *and* on test.

3 The rounds

Table 1 lists every variant. Rounds 1 and 2 fix false claims the baseline's templates made, found by the grader over all moments before the split was used to design anything. Rounds 3 to 9 add something the player asked for on the app's Coach Calibration page or that the judge's notes named, so the judge decides them.

Round 1: each line's material as it is.
The lesson's fallback sentence said "neither line wins material in the next 5 moves" whenever no earlier branch fired, including when both lines won the same material. It now says each line's material.

Round 2: the seventh-rank rook.
The insight named pawns the rook could not see past another pawn. It now names only the pawns the rook hits.

Round 3: what the better move defends.
A piece the reply takes, guarded or moved away by the better move (Figure 1).

Round 4: the capture a learner asks about first.
Where the better move or the reply lands on a square it can be taken on and the engine's line takes it, the lesson says what follows (Figure 2).

		test (%)		train (%)		texts	judge
variant	change	faithful	concrete	faithful	concrete	changed	train / test
baseline	the app before the rounds	90.1	75.0	87.6	78.0		
v1	Say each line's material as it is in the lesson's difference	96.1	82.2	98.3	84.7	20, 34	
v2	Name only the pawns the seventh-rank rook hits	100.0	82.2	99.4	84.7	25, 35	
v3	Say what the better move defends	100.0	82.2	99.4	84.7	31, 43	22–0 / 18–2
v4	Answer the capture a learner asks about first	100.0	82.9	99.4	84.7	34, 45	8–0 / 6–0
v5	Say where a long mate ends (*not kept*)	100.0	82.9	99.4	84.7	45, 54	6–6 / 14–2
v6	Say where a long mate ends, when the end is in reach	100.0	82.9	99.4	84.7	44, 51	4–0 / 10–0
v7	Say what a capture the better line declines would cost (*replaced by v7b*)	98.7	82.9	97.2	84.7	65, 77	48–3 / 40–1
v7b	Say what a capture wins, not the balance it leaves	100.0	82.9	99.4	84.7	65, 77	51–0 / 41–0
v8	Say when the played move is right but too early	100.0	82.9	99.4	84.7	70, 87	50–0 / 20–0
v9	Answer the natural capture, and skip what a pawn plainly guards (*not kept*)	100.0	82.2	98.9	85.9	142, 166	1–35 / 0–12

Table 1: Every variant. Rates are shares of key moments; *texts changed* counts the test and train moments whose visible texts differ from the baseline's; *judge* is the round's pooled preferences, new–old, on train and test (Table 2).

Rounds 5 and 6: where a long mate ends.

A mate is told with where it ends and what holds it there. Round 5 did so for every mate of three moves or more and failed its rule on train; round 6 keeps it for mates of three to seven moves with no promotion on the way.

Round 7: a capture the better line declines.

A search of each capture the line declines (`probes.mjs`, at the routine's budget), and the lesson says what it would cost: a recapture as the trade it is, or checks that win a piece, move by move (Figure 3). Round 7b fixes the material clause the judge found false in a few moments (a gain read as a balance).

Round 8: the right move, too early.

Where the move played is in the engine's best line at the mover's next or second turn, the difference and the takeaway say so, with the line (Figure 4).

Round 9 (not kept): fewer capture answers.

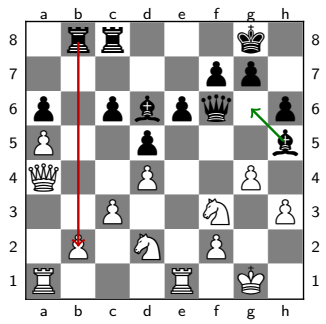
Skipping an answer when a pawn plainly guards the square, and stopping at the first capture with no answer. The judge wanted every answer back (Figure 5).

round	judged	train new–old (same)	test new–old (same)	the rule says
3	round 3	22–0 (0)	18–2 (0)	keep
4	round 4	8–0 (0)	6–0 (0)	keep
5	round 5	6–6 (10)	14–2 (12)	do not keep
6	round 5 (17 moments)	4–0 (8)	10–0 (12)	keep
7	round 7	48–3 (15)	40–1 (9)	keep
7b	round 7b	20–0 (0)	12–0 (0)	keep
7 and 7b	round 7 then round 7b	51–0 (15)	41–0 (9)	keep
8	round 8	50–0 (0)	20–0 (0)	keep
9	round 9	1–35 (0)	0–12 (0)	do not keep

Table 2: The judge rounds, pooled over both runs: new–old preferred, with the same-answers in parentheses. Round 6 made no run of its own: it reads round 5's judgments on the moments whose texts round 6 changes.

4 Examples

Each example is the first moment of its round, in the round's own order, on which both runs of the judge agreed. The tables show only the sections whose text the round changed.

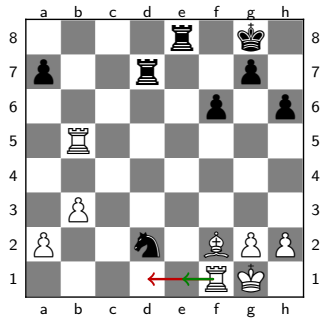


Played: 18. . . ♖xb2, Black to move; the app's verdict: Mistake.
Engine: +1.8 with the best move, +0.1 after yours.
Best line: 18. . . ♗g6 19. b4 h5 20. ♘e5 hxg4 21. ♘xg4
UCI: `b8b2`; best line `h5g6 b2b4 h6h5 f3e5 h5g4 e5g4`
FEN: `1rr3k1/5pp1/p1pbpq1p/P2p3b/Q2P2P1/2P2N1P/1P1N1P2/R3R1K1 b - - 0 18`

section	before	after
On the card	Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 avoids it. Mistake: it cost 15% of your winning chances.	Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 moves your bishop on h5 away, where it had no defender. Mistake: it cost 15% of your winning chances.
What Bg6 does instead	18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2	it moves your bishop on h5 away, where it had no defender · 18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2
The difference	After Rxb2, they win a pawn in the engine's line. Bg6 keeps it.	After Rxb2, they win a pawn in the engine's line. Bg6 keeps it: it moves your bishop on h5 away, where it had no defender.
Retry hint	*(not shown)*	Hint: it moves your bishop on h5 away, where it had no defender. The move starts on the lit square.
Retry answer	*(not shown)*	The move was Bg6. it moves your bishop on h5 away, where it had no defender

The judge (run 1, where the new text was Y): "Y says Bg6 rescues the attacked, undefended bishop on h5 that gxh5 wins, where X only says Bg6 'avoids it'."

Figure 1: Round 3: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

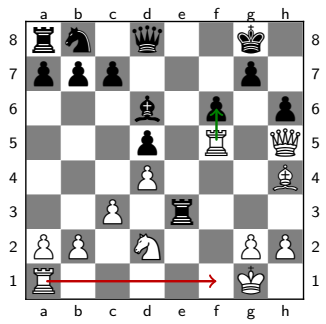


Played: 29. ♖d1, White to move; the app's verdict: Blunder.
Engine: +0.0 with the best move, −4.7 after yours.
Best line: 29. ♖e1 ♘e4 30. ♖b4 f5 31. ♖a4 ♖ee7
UCI: `f1d1`; best line `f1e1 d2e4 b5b4 f6f5 b4a4 e8e7`
FEN: `4r1k1/p2r2p1/5p1p/1R6/8/1P6/P2n1BPP/5RK1 w - - 1 29`

section	before	after
On the card	Rd1 allows Nf3+, and it attacks the pawn on h2. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances.	Rd1 allows Nf3+, and if you take it with gxf3, Rxd1+ takes your rook on d1. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances.
Takeaway	Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it attacks the pawn on h2.	Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it if you take it with gxf3, Rxd1+ takes your rook on d1.
What Rd1 allowed: Nf3+	it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3	if you take it with gxf3, Rxd1+ takes your rook on d1; it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3

The judge (run 1, where the new text was X): "X names the real tactic, that Nf3+ comes with check and uncovers the d7 rook so Rxd1+ wins your rook on d1, while Y only says the knight attacks h2, which does not explain the lost material."

Figure 2: Round 4: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

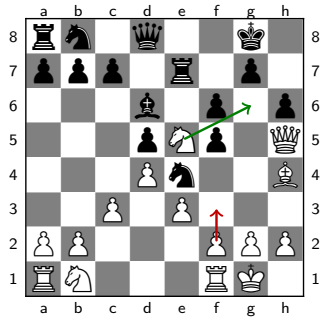


Played: 17. ♖af1, White to move; the app's verdict: Blunder.
Engine: +2.5 with the best move, 0.0 after yours.
Best line: 17. ♖xf6 ♕e8 18. ♕xd5+ ♔h8 19. ♖f7 ♘c6
UCI: `a1f1`; best line `f5f6 d8e8 h5d5 g8h8 f6f7 b8c6`
FEN: `rn1q2k1/ppp3p1/3b1p1p/3p1R1Q/3P3B/2P1r3/PP1N2PP/R5K1 w - - 0 17`

section	before	after
What Rxf6 does instead	it attacks the bishop on d6 and the pawn on h6 · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7	it attacks the bishop on d6 and the pawn on h6; if they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up; if they take it with Qxf6, Bxf6 wins their queen for your rook · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7
Retry answer	The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6.	The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6. If they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up. If they take it with Qxf6, Bxf6 wins their queen for your rook.

The judge (run 1, where the new text was X): "X shows why the rook sacrifice works (gxf6 Qg6+ Kf8 Qxh6+ then Qxe3 wins the rook back, and Qxf6 Bxf6 wins the queen), while Y leaves the player wondering why Rxf6 does not just lose a rook to gxf6."

Figure 3: Round 7: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

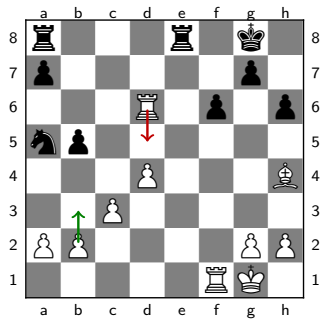


Played: 13. f3, White to move; the app's verdict: Inaccuracy.
Engine: +1.5 with the best move, +0.5 after yours.
Best line: 13. ♘g6 ♖e8 14. ♕xf5 c6 15. f3 ♘g5
UCI: `f2f3`; best line `e5g6 e7e8 h5f5 c7c6 f2f3 e4g5`
FEN: `rn1q2k1/ppp1r1p1/3b1p1p/3pNp1Q/3Pn2B/2P1P3/PP3PPP/RN3RK1 w - - 0 13`

section	before	after
Takeaway	Look for the more active move first: Ng6 attacks the rook on e7. f3 can often still be played a move later.	Look for the more active move first: Ng6 attacks the rook on e7. f3 still comes, two moves later, in the engine's line.
The difference	You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7.	You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7. The order matters: f3 comes in the engine's line too, two moves later (Ng6 Re8 Qxf5 c6 f3).

The judge (run 1, where the new text was X): "X says correctly that f3 still comes two moves later in the engine's line (Ng6 Re8 Qxf5 c6 f3), which shows the problem is move order; Y's generic 'can often still be played a move later' is vaguer and gets the count wrong."

Figure 4: Round 8: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.



Played: 23. ♖d5, White to move; the app's verdict: Inaccuracy.
Engine: +2.1 with the best move, +1.0 after yours.
Best line: 23. b3 ♖e2 24. ♖a6 ♘b7 25. ♖e1 ♖c2
UCI: `d6d5`; best line `b2b3 e8e2 d6a6 a5b7 f1e1 e2c2`
FEN: `r3r1k1/p5p1/3R1p1p/np6/3P3B/2P5/PP4PP/5RK1 w - - 1 23`

section	before	after
What b3 does instead	if they take it with Nxb3, axb3 wins their knight for your pawn · 23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8	23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8
Retry answer	The move was b3 (the green arrow). If they take it with Nxb3, axb3 wins their knight for your pawn.	The move was b3 (the green arrow).

The judge (run 1, where the new text was X): "Y adds the true safety check that b3 does not hang to the a5 knight, since Nxb3 axb3 wins the knight."

Figure 5: Round 9: the judge preferred the old text in both runs. The move played is the red arrow, the engine's best move the green one.

5 Results

From the baseline to the kept variant (v8), on the held-out games:

- faithful: 90.1 [84.2, 95.9]% to 100.0 [100.0, 100.0]%, a paired change of +9.9 [4.0, 15.9] points;
- concrete: 75.0 [65.3, 83.6]% to 82.9 [72.0, 91.4]%, a paired change of +7.9 [3.9, 12.1] points;
- named_deep: 94.1 [89.7, 97.9]% throughout, as expected of a property of the evidence;
- faithfulAll: 89.5 [83.3, 95.2]% to 99.3 [97.8, 100.0]%.

On train the same changes read 87.6% to 99.4% (faithful, +11.9 [7.9, 16.4]) and 78.0% to 84.7% (concrete, +6.8 [3.2, 10.6]). The kept variant changes the visible texts of 70 test and 87 train moments. Every kept round past the second was preferred by the judge on both halves; round 5 was not, and round 9 was rejected nearly unanimously.

6 Reproduction in the lab

Recomputed. The lab recomputes every variant's aggregates from the per-moment grades the records keep (`summarize.mjs`): 27 of the 27 lines of the published aggregates match exactly. Every judge round's tally is recomputed from the judge's answers and the order keys (`judge/pairwise.mjs`), and every decision follows from the rule as recorded. One recorded reason misstates its counts: round 5's reads "train 4 new vs 6 old (test 10 vs 2): the picture hurt mates of 8 to 13 moves", while the recorded answers give 6–6 on train and 14–2 on test. The new counts it gives are those of the moments round 6 later kept; the decision is the same, since a tie on train is not a majority.

Rerun. The lab rebuilt the engine evidence for the same games on the app as it now runs (`d8b7712`, evidence review-evidence/5), redid the capture searches on it (88 kept), and graded the app's texts. The rebuilt evidence is not the hill-climb's: the review pipeline changed after the rounds, so engine lines differ in detail and some key moments move (on test, 146 of 152 moments are the same moments; on train 171 of 177). The rates hold (Table 3). The records are read at `b3b827d`.

split	grade	recorded baseline	recorded v8	pinned app, rerun
test	faithful	90.1 [84.2, 95.9]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
test	concrete	75.0 [65.3, 83.6]	82.9 [72.0, 91.4]	82.9 [73.9, 90.5]
test	named_deep	94.1 [89.7, 97.9]	94.1 [89.7, 97.9]	94.1 [89.7, 97.8]
test	faithfulAll	89.5 [83.3, 95.2]	99.3 [97.8, 100.0]	99.3 [97.7, 100.0]
train	faithful	87.6 [83.0, 91.7]	99.4 [98.1, 100.0]	99.4 [98.1, 100.0]
train	concrete	78.0 [70.7, 84.2]	84.7 [78.9, 89.8]	83.6 [77.0, 89.4]
train	named_deep	92.1 [86.8, 97.1]	92.1 [86.8, 97.1]	92.1 [86.7, 97.2]
train	faithfulAll	87.6 [82.9, 91.8]	99.4 [98.2, 100.0]	99.4 [98.1, 100.0]

Table 3: The recorded baseline and kept variant beside the pinned app's texts on rebuilt evidence.

7 Limitations

- The grades check what can be checked: a true and concrete text can still fail to teach. That question goes to the player on the Calibration page, and to the judge, whose agreement with the player is itself only partly measured.

- The judge is a model reading on the player's behalf. It is a good finder of what is missing and a poorer judge of how much it matters, which is why the rounds use its pairwise preferences and its notes, not its ratings.
- 60 games of one player. The split keeps the claims honest about that player, not about players in general.